

JKBOSE 2026

CLASS 11TH

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ON
TOP.

#XUV ❤️

HYDROCARBONS

LECTURE-02

CHEMISTRY

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Topics To Covered

- 1** Conformation of Ethane
- 2** Preparation of Alkanes
- 3** Physical Properties of Alkanes
- 4** Chemical Properties of Alkanes

REVISION.

ISOMERISM.

STRUCTURAL.

- 1) SKELETON OR CHAIN.
- 2) POSITION
- 3) Functional group
- 4) Metamerism

STEREoisomerism.

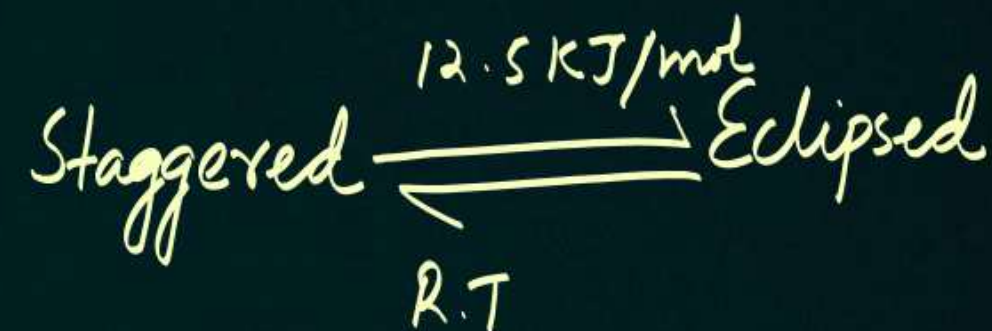
- a) Geometrical.
- b) OPTICAL.

QUESTION



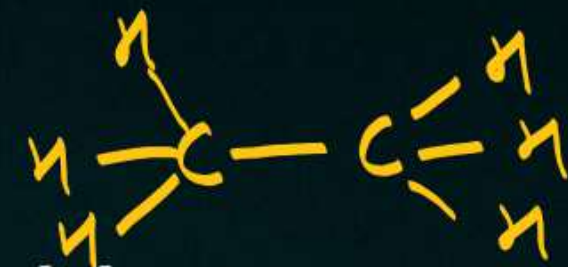
Conformation of ethane in which hydrogen atoms attached to two carbon atoms are closest to each other are known as.

- 1 Eclipsed
- 2 Skew boat
- 3 Staggered
- 4 Partially eclipsed



Conformation of Ethane

Important



- The different arrangements of atoms in space which can be obtained due to rotation about carbon-carbon single bond are called conformations.



Free rotation around
C – C σ bond

- These different spatial arrangements are called conformational isomers (conformers) or rotational isomers (rotamers). Thus,



Out of numerous possible arrangements we usually study two extreme cases.

a) Eclipsed : $\rightarrow$ In this conformation two hydrogen atoms present on adjacent carbon atoms are as close as possible.

b) STAGGERED CONFORMATION : $\rightarrow$ In this conformation two hydrogens present on adjacent carbon are as far as possible.

These two 3D conformations can be represented in 2D by two ways.

1) SAWHORSE FORMULA.

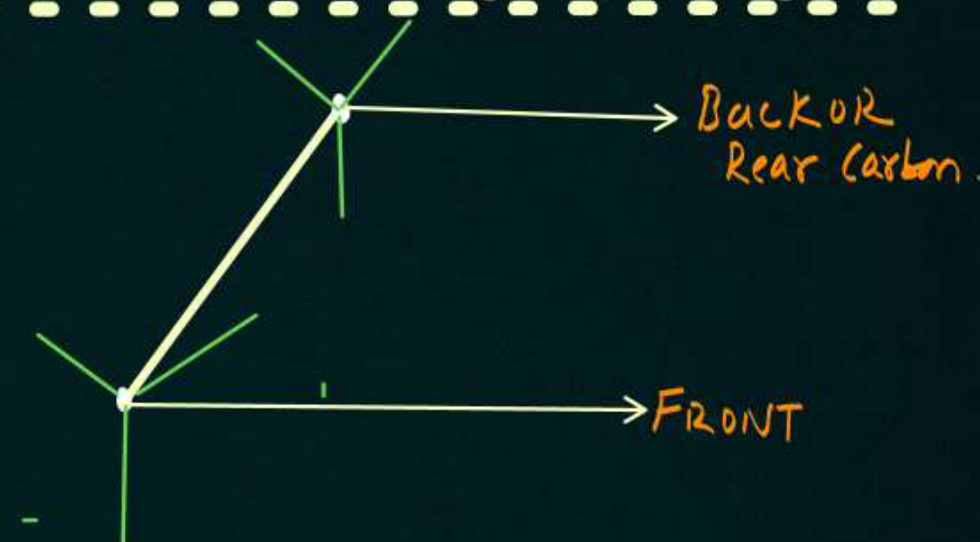
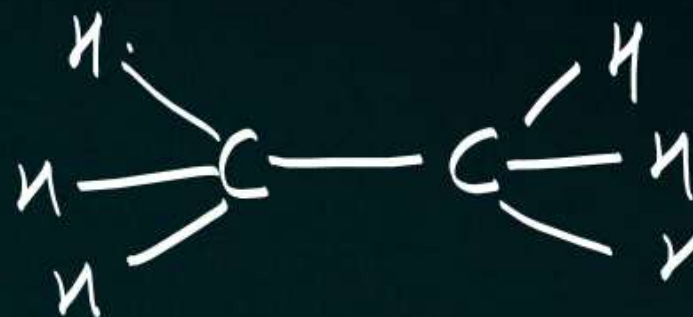
2) NEWMANN PROJECTION.

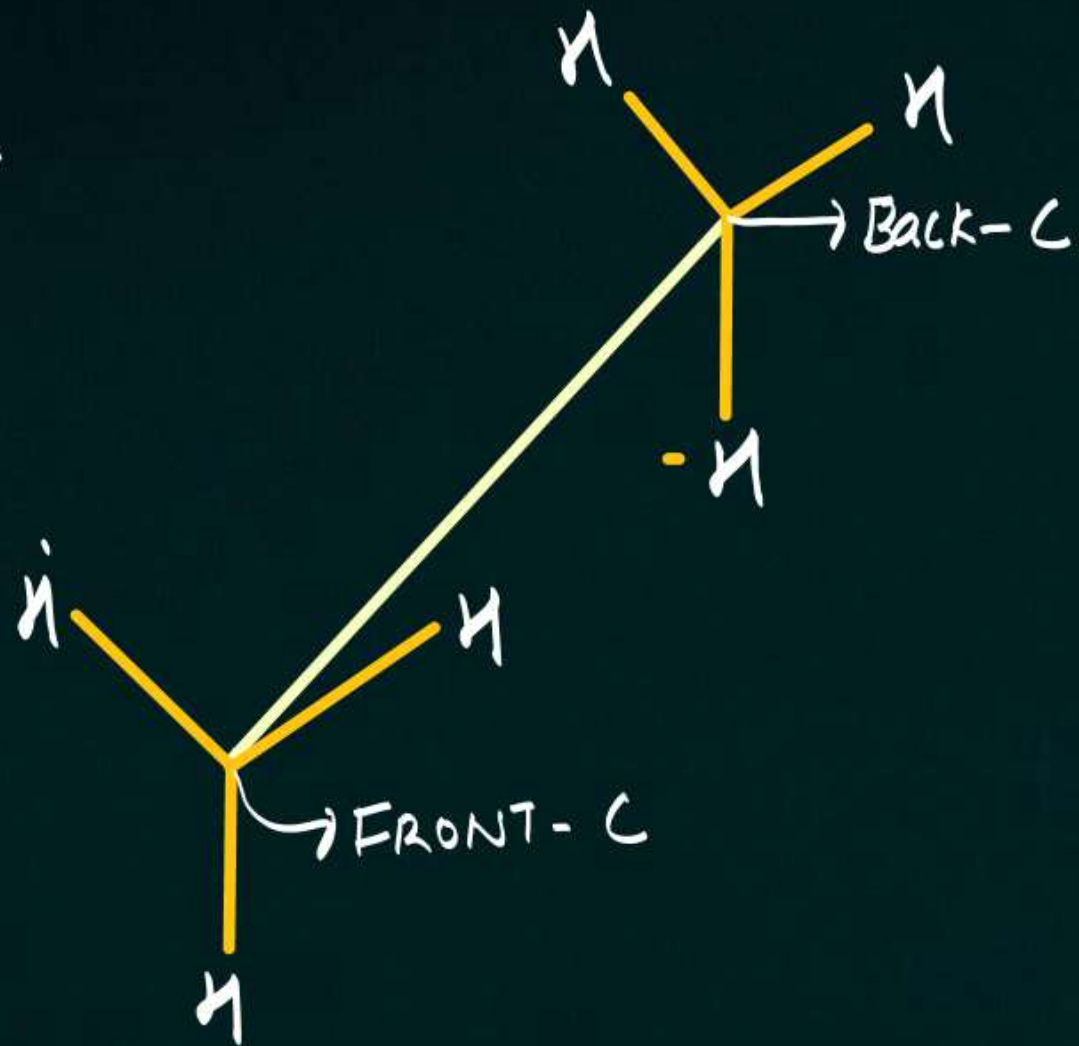
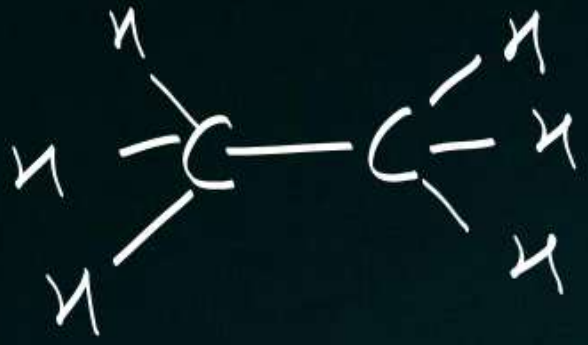
Conformation of Ethane



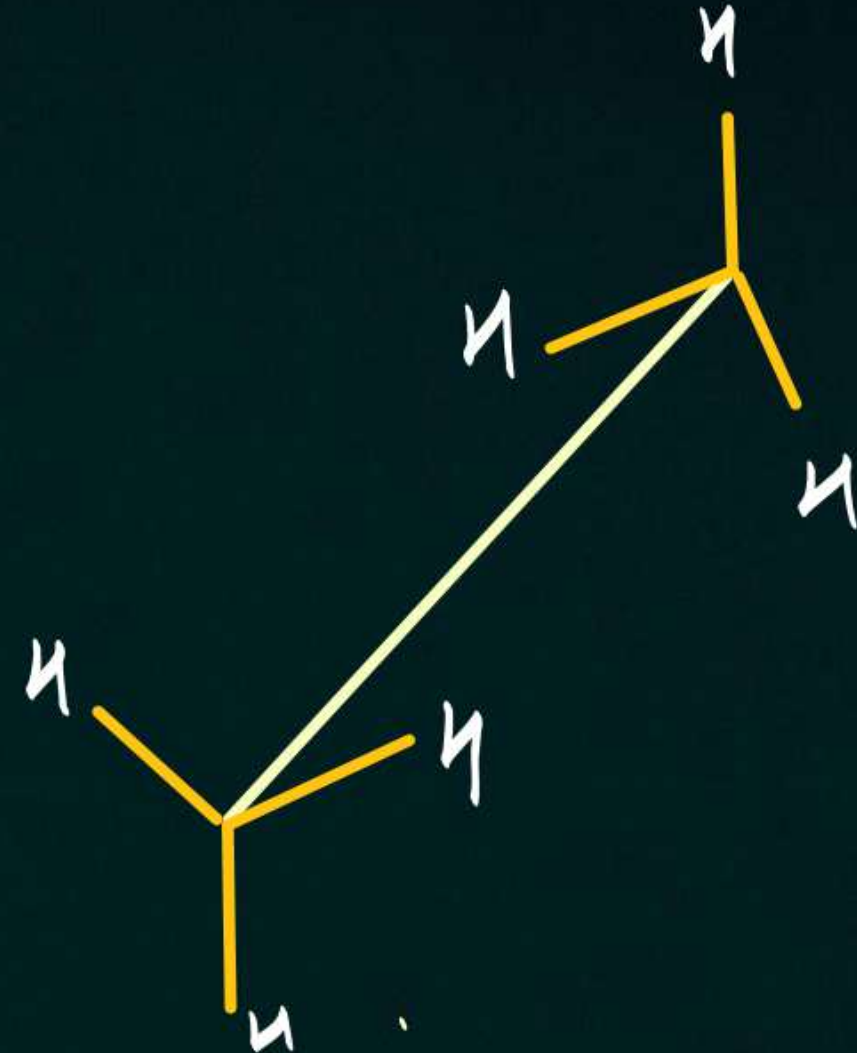
(A) Sawhorse Representation:

- **Central Bond Representation:** The central C–C bond is drawn as a slightly longer straight line, tilted slightly to the right or left.
- **Carbon Placement:**
Front carbon: Shown at the lower end of the line.
Rear carbon: Shown at the upper end of the line.
- **Attached Atoms:** Each carbon is connected to three lines (representing three atoms) inclined at an angle of 120° to each other.
- **Conformations:** Sawhorse projections represent both eclipsed and staggered conformations of ethane.



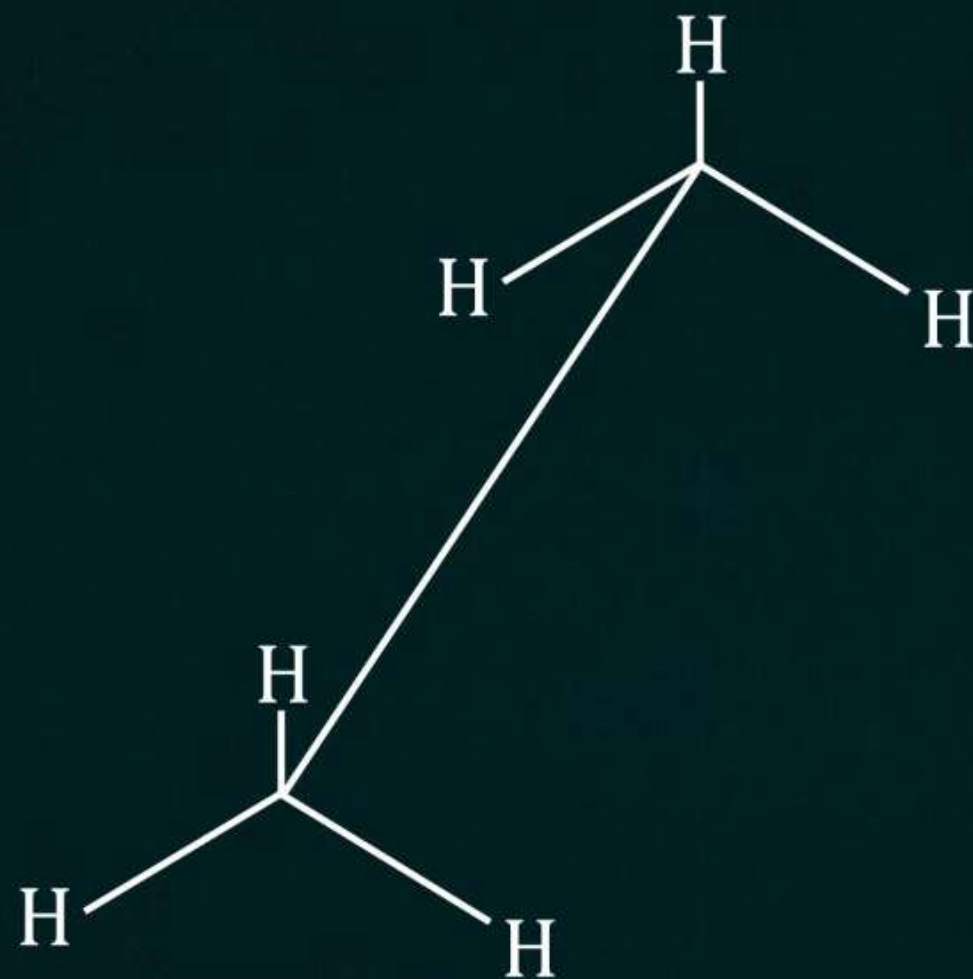


ECLIPSED

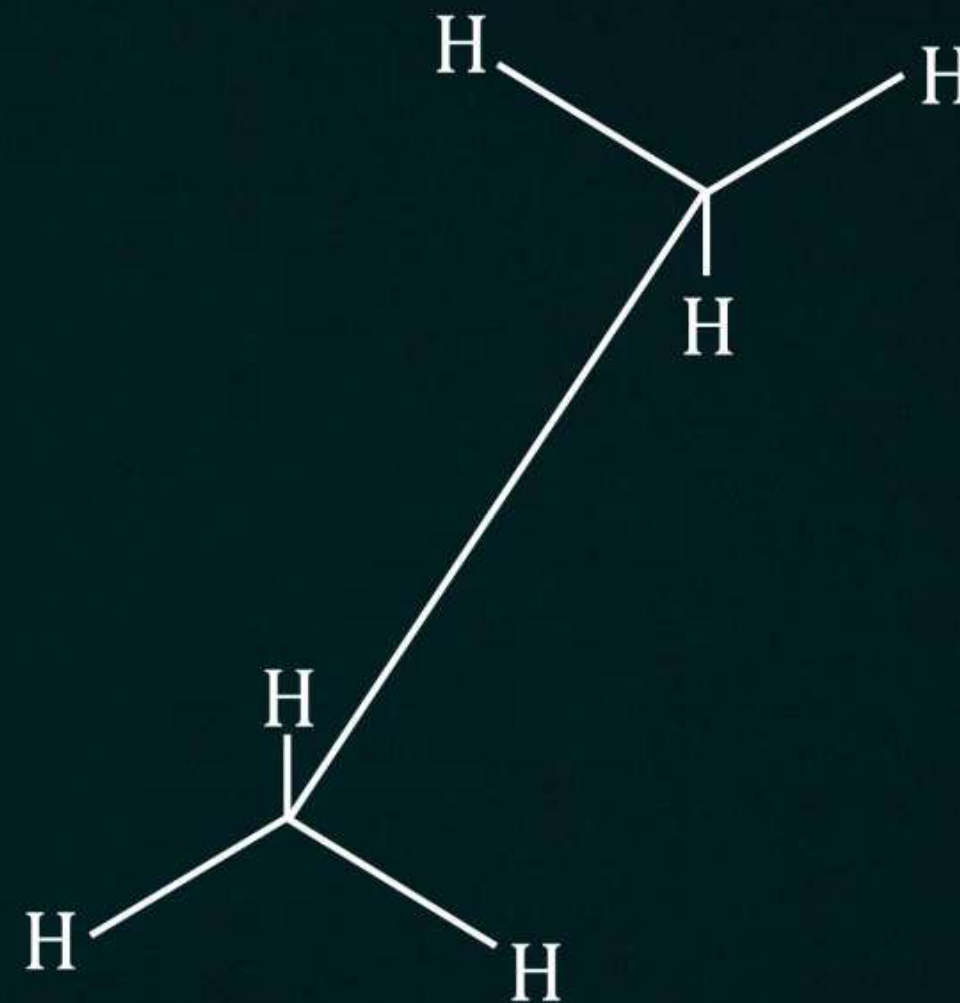


STAGGERED

Conformation of Ethane



(i) Eclipsed



(ii) Staggered

Sawhorse projections of ethane

Conformation of Ethane

(A) Newman Projection:

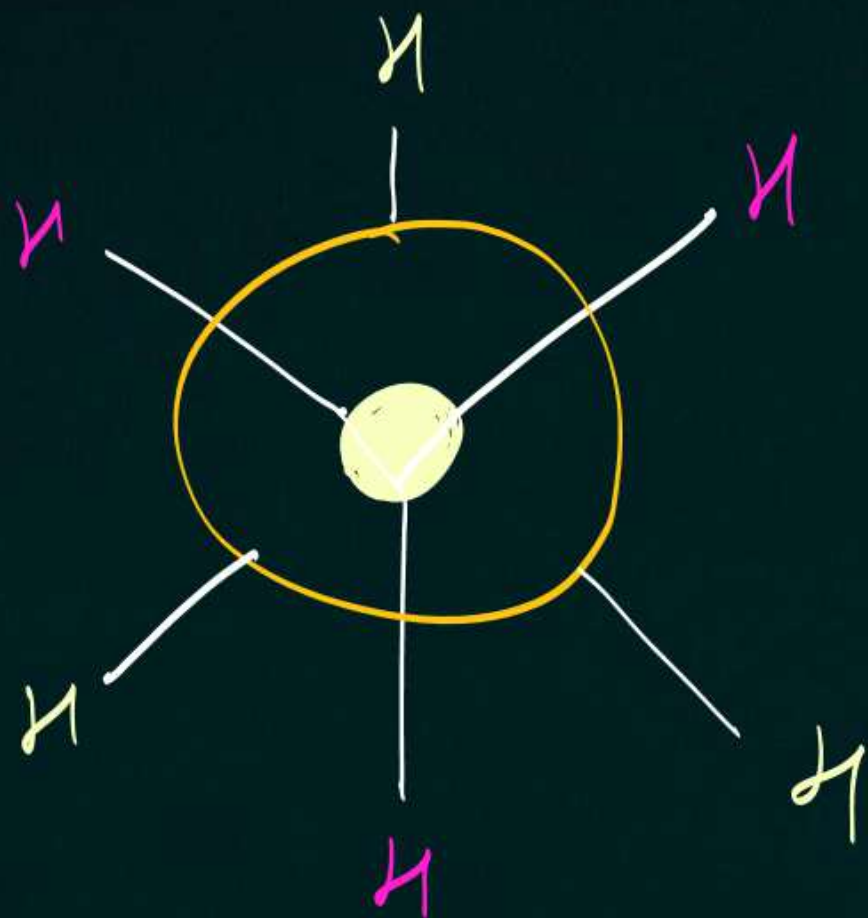
- **Viewpoint:** Molecule is observed head-on along the C-C bond axis.
- **Front Carbon Representation:**
Shown as a point.
Three hydrogen atoms are attached with lines at 120° angles.
- **Rear Carbon Representation:**
Shown as a circle behind the front carbon.
Three hydrogen atoms are attached with shorter lines at 120° angles.



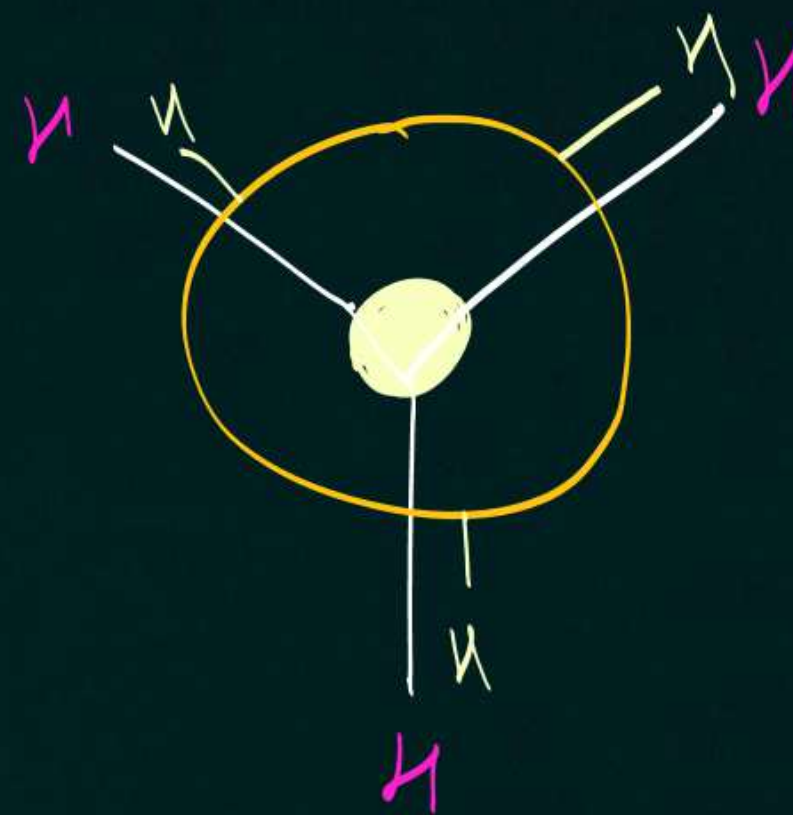
ECLIPSED.



STAGGERED

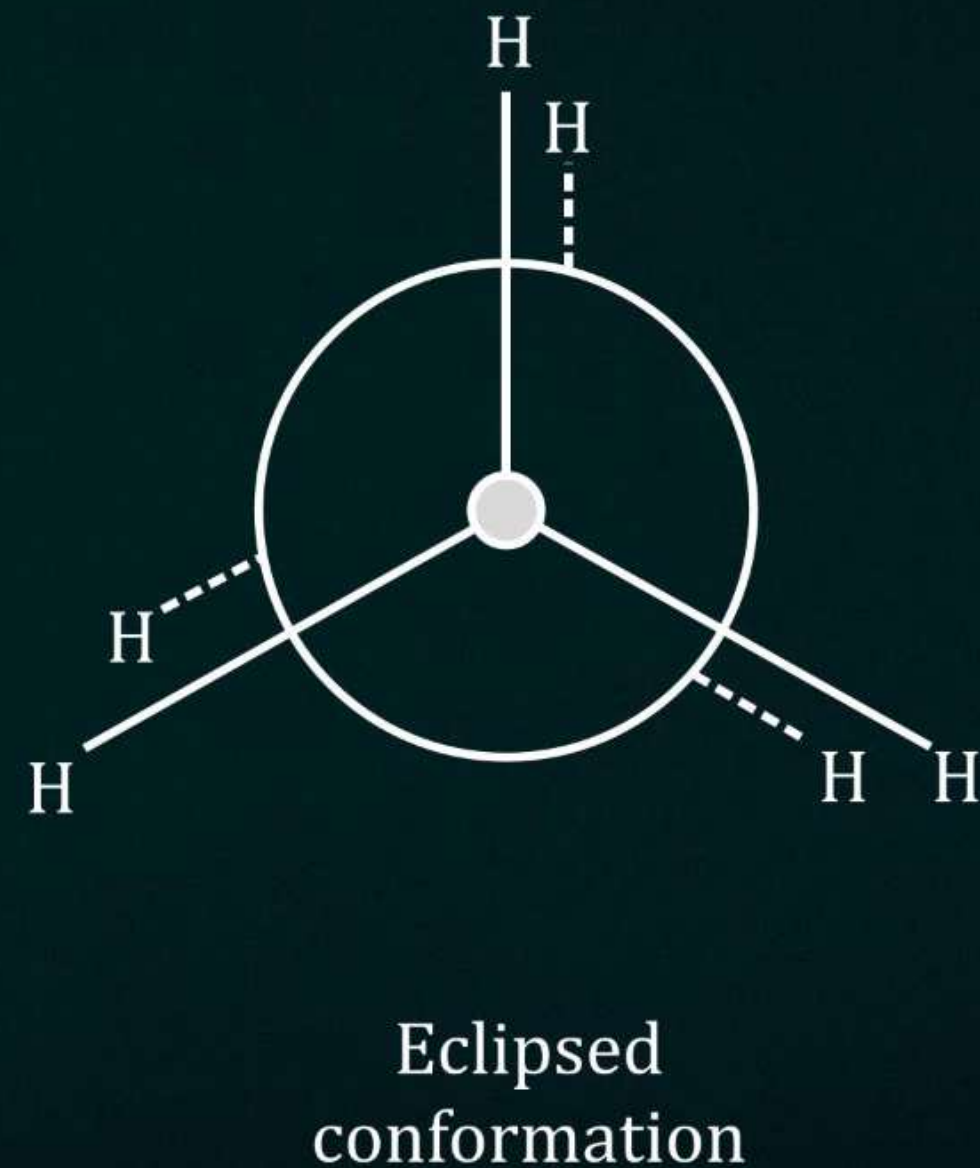
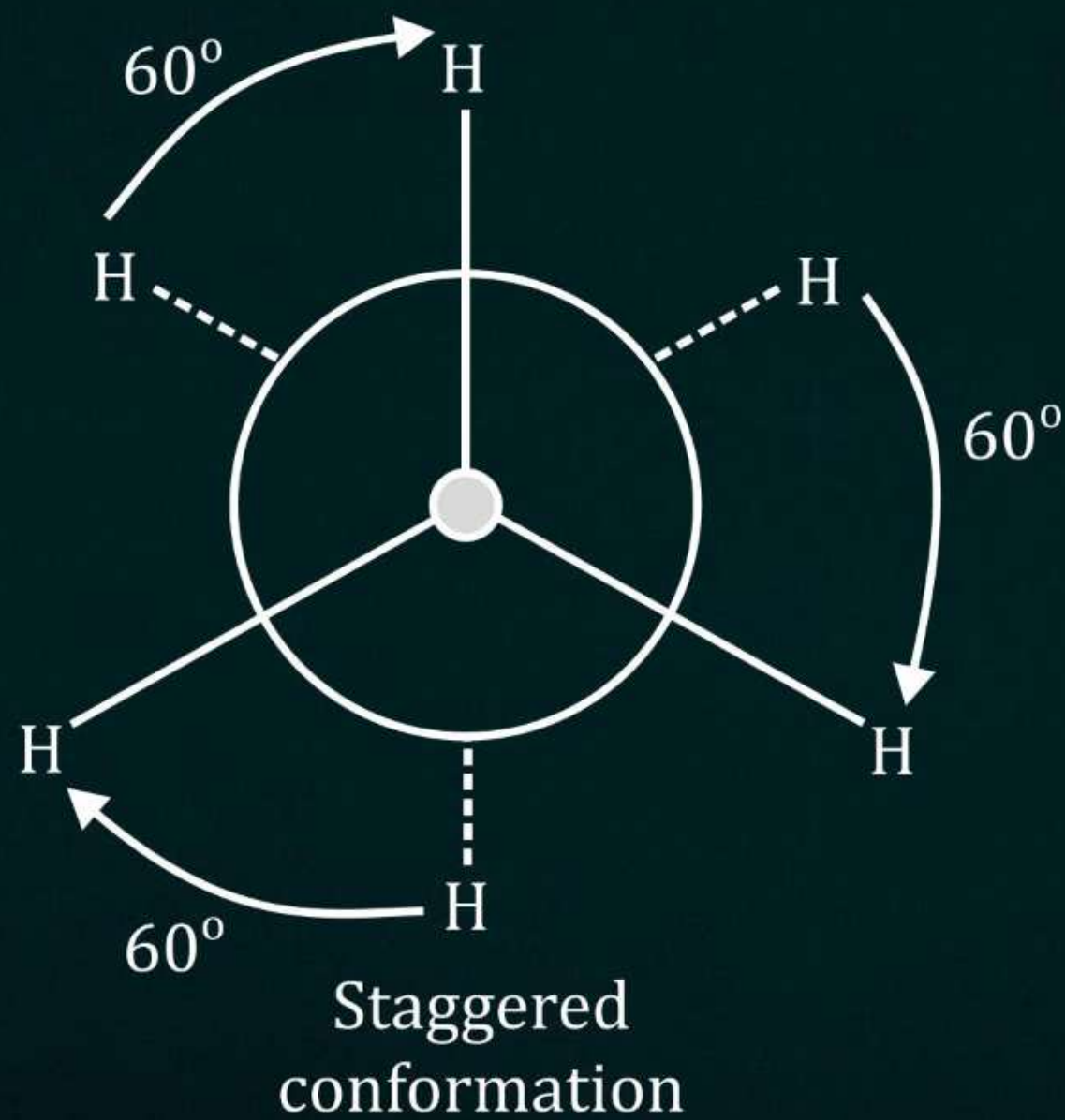


STAGGERED.



ECLIPSED.

Conformation of Ethane



Relative Stability of Eclipsed and Staggered Conformation

1. Staggered conformation is more stable due to following reasons.

→ Minimum repulsion between sigma $\bar{e}$ of front and rear $-C-H$ bond.

→ Minimum energy of molecule therefore maximum stability.

(Repulsion $\downarrow$ Energy $\downarrow$ Stability $\uparrow$)

2. Eclipsed conformation is less stable due to following reasons

→ Maximum repulsion of sigma $\bar{e}$ of front and rear $C-H$ bond

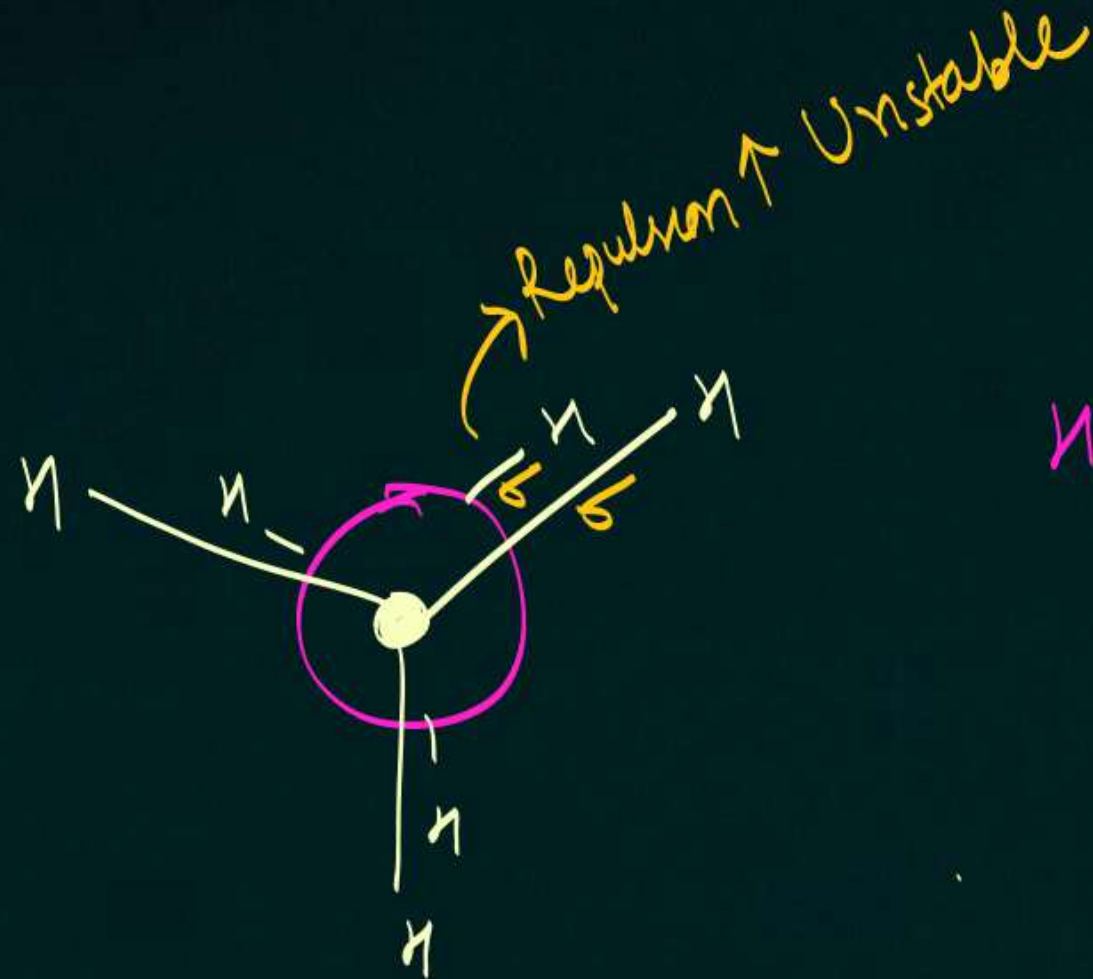
→ Maximum energy of molecule therefore minimum stability.

(Repulsion $\uparrow$ Energy $\uparrow$ Stability $\downarrow$)

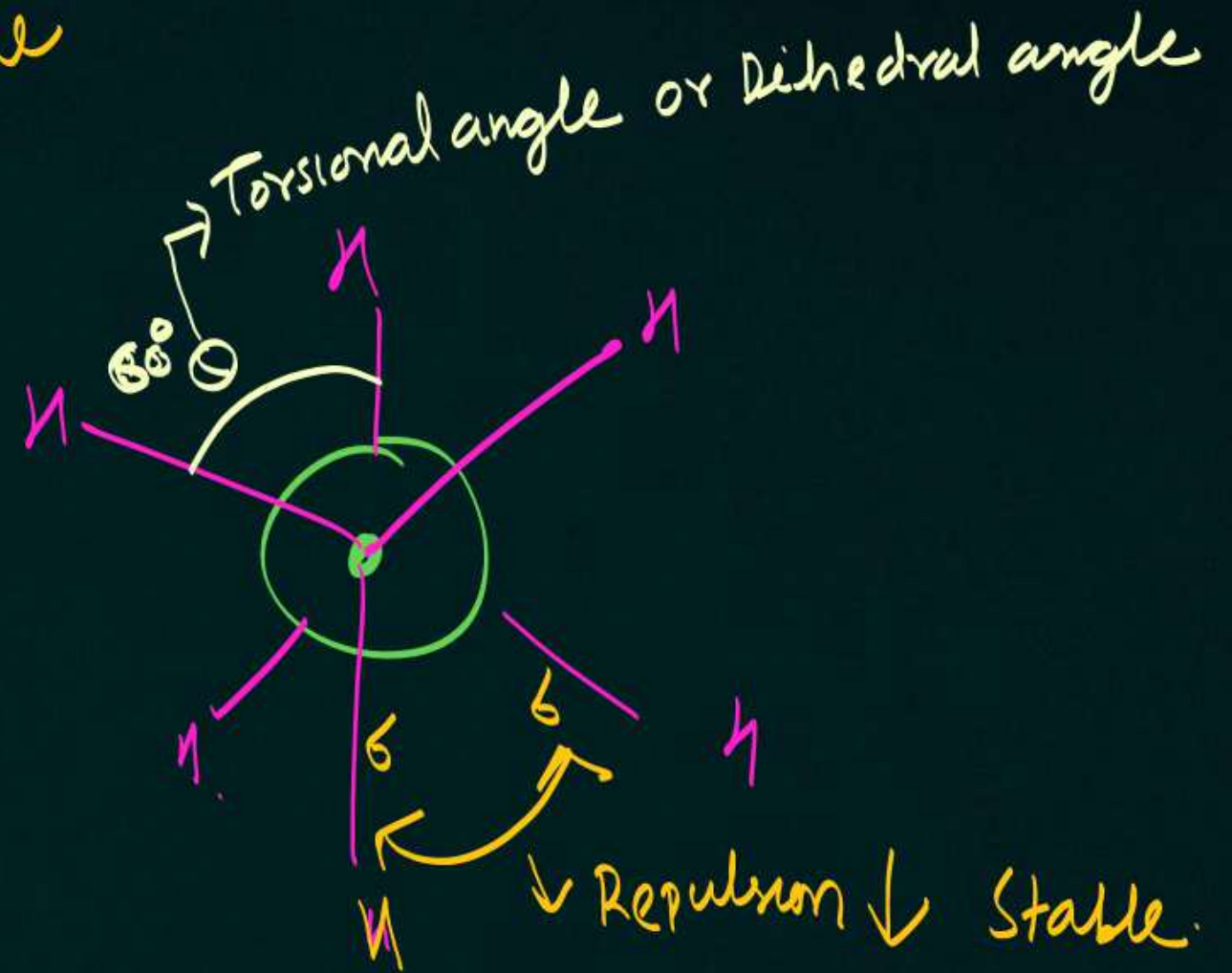
3. Torsional Strain: It is the repulsion between the sigma $\bar{e}$ of front and rear $C-H$ bond.

The angle present ^{present between} $C-H$ bond of front and rear carbon is called torsional angle.

This torsional angle maximum in case of staggered & minimum in case of eclipsed conformation.



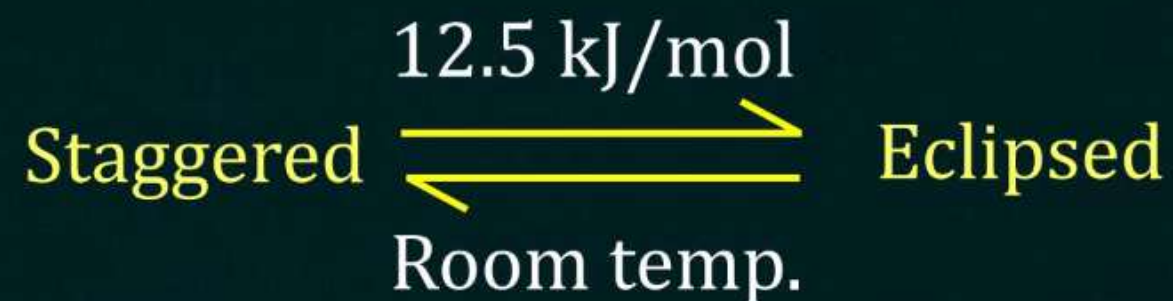
ECLIPSED



STAGGERED

Relative Stability of Eclipsed and Staggered Conformation

4. Preferred Conformation: Ethane molecule prefers staggered conformation because it is more stable. But at room temperature both the conformation exist.



QUESTION

Eclipsed and the staggered conformations can be represented by

- 1 Sawhorse projection
- 2 Newman projection
- 3 Both (1) and (2)
- 4 None of the above

QUESTION



How many chain isomers can be formed by butane?

- 1 One
- 2 Three
- 3 Two
- 4 Four

QUESTION



The given structures,



1 Conformational isomers

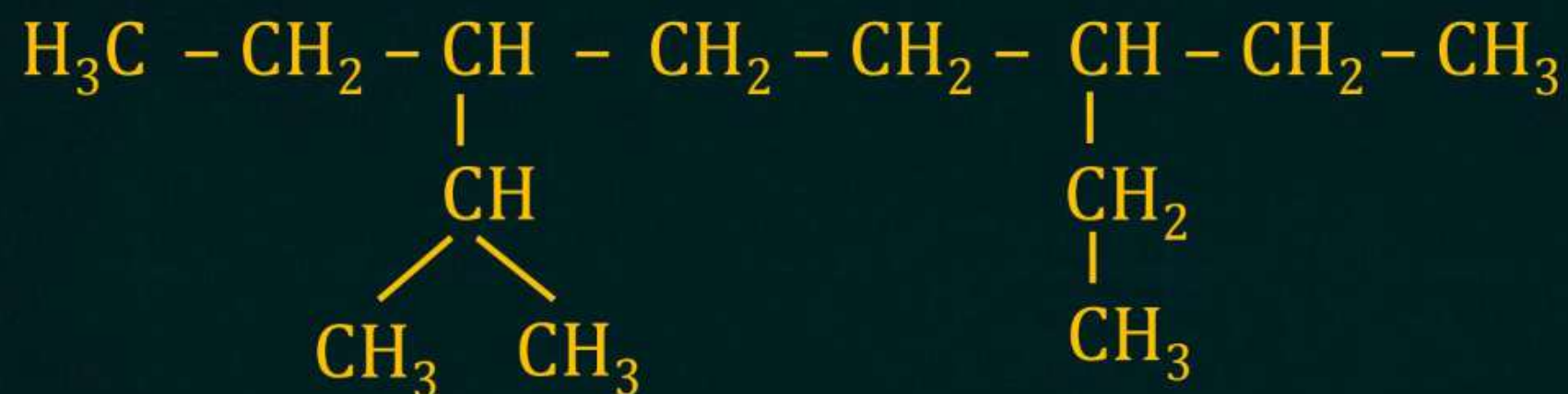
2 Chain isomers

3 Position isomers

4 Functional isomers

QUESTION

The correct IUPAC name of the following alkane is



- | | |
|-------------------------------------|-------------------------------------|
| 1 3,6-diethyl-2-methyloctane | 2 5-isopropyl -3-ethyloctane |
| 3 3-ethyl-5-isopropyloctane | 4 3-isopropyl-6-ethyloctane |

Thank
You